

AAKRITI BARAT

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[Web Portfolio](#) • Athens, OH • Open to Relocation Nationwide

PROFESSIONAL PROFILE

Hardware security researcher and engineer with PhD-level expertise in silicon-level trust primitives, Physically Unclonable Functions (PUF), and IC authentication, combined with 3+ years of industry experience spanning PCB design, embedded systems, and hardware validation. Designs and validates trust primitives from first principles — TCAD/HSPICE device simulation through direct silicon probing. First author on 5 peer-reviewed IEEE and ACM publications on supply-chain security, TEE-aligned authentication, IC counterfeiting prevention, and device reliability. Experienced teacher, mentor, and workshop organizer with a strong record of translating advanced hardware security research into accessible, hands-on learning. Eligible for cap-exempt H-1B sponsorship at qualifying research institutions (national labs, nonprofits, universities) and cap-subject H-1B for industry.

RESEARCH INTERESTS

Physically Unclonable Functions (PUF) & entropy characterization • Root of Trust architecture • IC authentication & supply-chain security • Device-level security modeling (TCAD/HSPICE) • Trusted Execution Environment (TEE) design principles • Side-channel analysis & fault injection countermeasures • Current Transient Spectroscopy (CTS) for process authentication • Cyber-physical system resilience • Bayesian inference for hardware characterization • Secure embedded systems

TECHNICAL SKILLS

Hardware Security: Physically Unclonable Function (PUF) • Root of Trust • Secure Boot • Hardware Cryptography • Side-Channel Analysis • Fault Injection • JTAG Security • IC Authentication • Supply-Chain Security • TEE Design Principles

Simulation & Modeling: TCAD (device-level) • HSPICE (circuit-level) • Python • MATLAB • Bayesian Inference • Multi-Exponential Signal Processing • Statistical Analysis & Visualization

Experimental: CMOS Device Probing & Characterization • Current Transient Spectroscopy (CTS) • Silicon Validation • Correlation Analysis • Prototype Fabrication & Testing

FPGA & RTL: VHDL • RTL Design • FPGA (Xilinx / Lattice ECP3) • Mentor Graphics

PCB & Hardware: Schematic Capture • PCB Layout • BOM • DFM • DFT • Prototype Fabrication • Signal Analysis • Root-Cause Analysis

Embedded Systems: C/C++ • CAN Bus • UART • SPI • I2C • Interrupt-Driven Firmware • Motor Control • 16-bit ADC • Arduino • Raspberry Pi • Real-Time Systems

Tools & Workflows: Git • Jupyter • AWS (Data Analytics) • Agile PLM • Bash • ECN/ECO Workflows • Requirements Documentation • Risk Analysis

PUBLICATIONS

[1] A. Barat et al.. "Design and Verification of a Novel PUF for Enhanced IC Security Based on Total Harmonic Distortion in CMOS Inverters." IEEE MWSCAS, 2025. **(First Author)**

[2] A. Barat et al.. "On-Chip Monitoring of Traps via Current Transient Spectroscopy: A New Security Tool for Process Authentication and Device Reliability." IEEE MWSCAS, 2025. **(First Author)**

- [3] A. Barat et al.. "A PUF-Based Security Framework for Fault and Intrusion Detection." IEEE ICCE, 2026. **(First Author)**
- [4] A. Barat et al.. "Process Identification and PUF Design Using CMOS Inverter Nonlinearity: Demonstration via 45nm & 22nm CMOS." IEEE/ACM GLSVLSI, 2026. **(First Author)**
- [5] A. Barat et al.. "Hardware Validation of CMOS Inverter PUF Using Commercial Integrated Circuits." IEEE MWSCAS, 2026. **(First Author)**

Active peer reviewer: ACM and IEEE (hardware security, VLSI, cryptographic engineering, device reliability).

RESEARCH EXPERIENCE

PhD Researcher — Hardware Security & On-Chip Authentication

Defense: Fall 2026

Ohio University, Dept. of Electrical Engineering & Computer Science

Athens, OH

- Designed a novel CMOS inverter-based Physically Unclonable Function (PUF) leveraging Total Harmonic Distortion (THD) as entropy source; demonstrated scalability at 45nm and 22nm CMOS nodes and validated PUF response on commercial integrated circuits through hardware probing.
- Validated PUF architecture through rigorous HSPICE circuit-level and TCAD device-level simulations; computed entropy, uniqueness, and reliability metrics across process corners and PVT (process, voltage, temperature) operating conditions.
- Performed direct CMOS device characterization via experimental silicon probing, correlating measured Current Transient Spectroscopy (CTS) data with TCAD trap models to develop a new on-chip security tool for process authentication and device reliability monitoring.
- Built Python/MATLAB statistical pipelines for large-scale simulation dataset processing, applying Bayesian inference and multi-exponential signal decomposition to extract security-relevant device parameters.
- Co-developed PUF-based hardware fault and intrusion detection frameworks targeting TEE-aligned authentication, IC counterfeiting prevention, and aging/reliability monitoring; contributed to 5 peer-reviewed IEEE and ACM publications (first author on all).
- Secured \$15,000 departmental research funding; organized and delivered 3-day hardware security workshop training 25 K-12 teachers and students.

Embedded Systems Research Associate — FPGA & Thermal Imaging

Aug 2014 – Jul 2015

Indian Institute of Technology, Madras (IIT Madras)

India

- Engineered a multichannel 3-sub-aperture FPGA-based aerial thermal imaging system; developed a VHDL 3x3 windowing module on Lattice ECP3-70EA FPGA achieving 50% resolution improvement (to 960x720 output) across 5 prototypes.
- Delivered a sub-100g, 5V/1A thermal camera validated across 3 nighttime field missions, integrating FPGA-based HDMI display, Python preprocessing pipeline (95% image size reduction via antialiasing), and a 4-board PCB assembly.

Graduate Research Associate — Cleanroom & Nano-Fabrication

Aug 2015 – Jan 2017

The Ohio State University

Columbus, OH

- Conducted research on Phase Locked Loops and wavelet analysis; performed cleanroom nano-fabrication work as part of MS thesis in Electrical & Computer Engineering.

INDUSTRY & INTERNSHIP EXPERIENCE

Embedded Systems & Controls Engineering Intern

May 2024 – Aug 2024

OsteoDx, Inc.

Athens, OH

- Engineered a CAN-based multi-motor control system (Arduino Uno + MCP2515) achieving 500 Kbps throughput and 10 Hz synchronization across 3 BLDC motors via Timer1 interrupt-driven firmware; reduced motor positioning latency by 20%.
- Cut telemetry data loss by 30% (ACK/NACK protocol verified across 9,644 trials) via MATLAB-Arduino handshake pipeline with real-time interrupt-driven firmware.
- Delivered 99% sensor reliability across 22,740 clinical trials (zero hardware failures) integrating 16-bit ADC with trapezoidal motion profiles and Emergency Stop.
- Achieved >97% diagnostic sensitivity/specificity across 758 bilateral patient datasets via Python telemetry extraction and iterative signal-to-noise quality scoring.

Senior Electrical Hardware Design Engineer

Dec 2017 – Mar 2020

Vertiv

Columbus, OH

- Led end-to-end PCB design and validation for server infrastructure platforms — requirements specification through prototype fabrication and manufacturing handoff.
- Developed global testing protocol adopted by international contract manufacturers, expediting production testing by 80% and reducing variability across sites.
- Performed root-cause analysis of field returns; implemented corrective measures preventing data loss in deployed server rack products.
- Supported cross-functional team of 10+ (engineering, product, sales) through full PCB lifecycle, risk analysis documentation, and server rack product launch.

Electrical Hardware Engineer

Mar 2017 – Dec 2017

Thermo Fisher Scientific

Cleveland, OH

- Re-engineered 15+ legacy radiation measurement instruments, saving \$500K in redesign costs and extending product lifespan by 5 years; redesigned 300+ schematics using Agile PLM.
- Raised manufacturing productivity by 40% (Agile PLM audit verified); processed 50+ ECNs weekly; grew quarterly profit margins by 15% through 20+ vendor relationships.
- Promoted to youngest engineer within 2 months based on performance and impact.

TEACHING & MENTORSHIP

Graduate Teaching Assistant

Aug 2021 – May 2025

Ohio University, Dept. of Electrical Engineering & Computer Science

Athens, OH

- Embedded Systems (EE 4900/5900): Designed lab curriculum; facilitated FPGA, Arduino, CAN bus projects — 95% project completion rate.
- Hardware Security (Cybersecurity Bachelor's Track): Co-instructed 50+ students per semester; 100% course pass rate across all cohorts.
- Digital Circuits: Delivered FPGA and VHDL instruction; graded labs and examinations.
- Micro & Nanofabrication / Nano-electronic Devices: Supported lab instruction and student assessment in fabrication concepts.
- Mentored junior PhD and MS researchers in TCAD/HSPICE workflows, experimental setup, data analysis, and academic writing.

OUTREACH, LEADERSHIP & SERVICE

- SSEP Facilitator — NASA SpaceX Falcon 9 Mission 18 ISS Payload (2022–2023): Led undergraduate teams designing a flight experiment for the International Space Station meeting rigorous NASA standards; coordinated \$15,000 departmental funding.
- Hardware Security Summer Workshop Organizer (2021–2023, 2025): Organized and led 3-day workshops for 25 K-12 teachers and students using Raspberry Pi, Jupyter notebooks, and hardware hacking platforms; adapted curriculum for both teacher PD and youth audiences.
- Peer reviewer for ACM and IEEE: hardware security, VLSI, cryptographic engineering, and device reliability (ongoing, multiple venues 2024–2026).

EDUCATION

Doctor of Philosophy (PhD), Electrical Engineering & Computer Science

Defense: Fall 2026

Ohio University, Athens, OH — Research: Hardware Security, PUF, On-Chip Authentication

Master of Science (MS), Electrical & Computer Engineering

Aug 2015 – Jan 2017

The Ohio State University, Columbus, OH — Thesis: Analysis and Design of Phase Locked Loops with Insight into Wavelet Analysis

Bachelor of Engineering (BE), Electronics & Communication Engineering

2010 – 2014

University Institute of Technology, RGPV, India

HONORS & AWARDS

- Dr. Harold F. Klock Endowed Scholarship — Ohio University (outstanding research performance)
- G.E. Smith and G.V. Smith Memorial Award — Ohio University
- Promoted to youngest engineer within 2 months of hire — Thermo Fisher Scientific
- SSEP Facilitator — NASA SpaceX Falcon 9 Mission 18 ISS Payload (2022–2023)